

Implementing TLS, SSO, and Vault with Keycloak

1. Showing keycloak is already installed

```
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4
hikmat@hikmat-VMware-Virtual-Platform:~$ tar -xzf keycloak-26.3.4.tar.gz
hikmat@hikmat-VMware-Virtual-Platform:~$ cd keycloak-26.3.4
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ ls bin/kc.sh
bin/kc.sh
```

2. Creating a self-signed cert (with SAN) inside the Keycloak folder to work for <https://localhost>.

```

hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ ls bin/kc.sh
bin/kc.sh
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ mkdir -p ~/keycloak-26.3.4/certs

openssl req -x509 -newkey rsa:2048 -nodes -days 365 \
-keyout ~/keycloak-26.3.4/certs/keycloak.key \
-out ~/keycloak-26.3.4/certs/keycloak.crt \
-subj "/CN=localhost" \
-addext "subjectAltName=DNS:localhost,IP:127.0.0.1"

```

3. Configuration forces HTTPS

```
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ ls ~/keycloak-26.3.4/certs/
# Should show: keycloak.crt  keycloak.key
keycloak.crt  keycloak.key
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ mkdir -p ~/keycloak-26.3.4/conf

cat > ~/keycloak-26.3.4/conf/keycloak.conf <<'CONF'
https-port=8443
http-enabled=false
https-certificate-file=${kc.home.dir}/certs/keycloak.crt
https-certificate-key-file=${kc.home.dir}/certs/keycloak.key
hostname=localhost
hostname-strict=false
CONF
```

4. Starting Keycloak

```

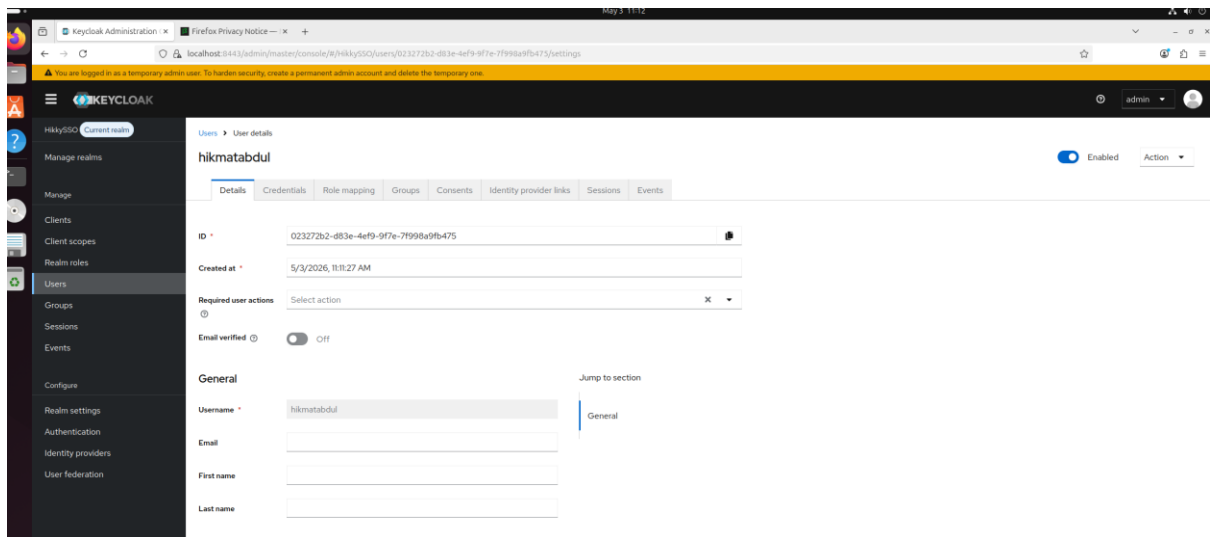
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ cd ~/keycloak-26.3.4
bin/kc.sh start-dev
Updating the configuration and installing your custom providers, if any. Please wait.
2026-05-03 10:28:53,200 INFO [io.quarkus.deployment.QuarkusAugmentor] (main) Quarkus augmentation completed in 22502ms
Running the server in development mode. DO NOT use this configuration in production.
2026-05-03 10:28:58,990 INFO [org.keycloak.url.HostnameV2ProviderFactory] (main) If hostname is specified, hostname-strict is effectively i
2026-05-03 10:29:07,449 INFO [org.keycloak.quarkus.runtime.storage.database.liquibase.QuarkusJpaUpdaterProvider] (main) Initializing databa
2026-05-03 10:29:13,631 INFO [org.keycloak.spi.infinispan.impl.embedded.JGroupsConfigurator] (main) JGroups JDBC_PING discovery enabled.
2026-05-03 10:29:13,774 INFO [org.infinispan.CONTAINER] (main) Virtual threads support enabled
2026-05-03 10:29:13,965 INFO [org.infinispan.CONTAINER] (main) ISPN000556: Starting user marshaller 'org.infinispan.commons.marshall.Immutable
2026-05-03 10:29:14,375 INFO [org.keycloak.connections.infinispan.DefaultInfinispanConnectionFactory] (main) Node name: node_230029
2026-05-03 10:29:14,530 INFO [org.keycloak.services] (main) KC-SERVICES0050: Initializing master realm
2026-05-03 10:29:17,067 WARN [org.keycloak.services] (main) KC-SERVICES0110: Environment variable 'KEYCLOAK_ADMIN' is deprecated, use 'KC_E
2026-05-03 10:29:17,069 WARN [org.keycloak.services] (main) KC-SERVICES0110: Environment variable 'KEYCLOAK_ADMIN_PASSWORD' is deprecated,
2026-05-03 10:29:17,294 INFO [org.keycloak.services] (main) KC-SERVICES0077: Created temporary admin user with username admin
2026-05-03 10:29:17,583 INFO [io.quarkus] (main) Keycloak 26.3.4 on JVM (powered by Quarkus 3.20.2.2) started in 23.987s. Listening on: htt
2026-05-03 10:29:17,586 INFO [io.quarkus] (main) Profile dev activated.
2026-05-03 10:29:17,586 INFO [io.quarkus] (main) Installed features: [agroal, cdi, hibernate-orm, jdbc-h2, keycloak, narayana-jta, opentele

```

5. Keycloak listening on https://localhost:8443

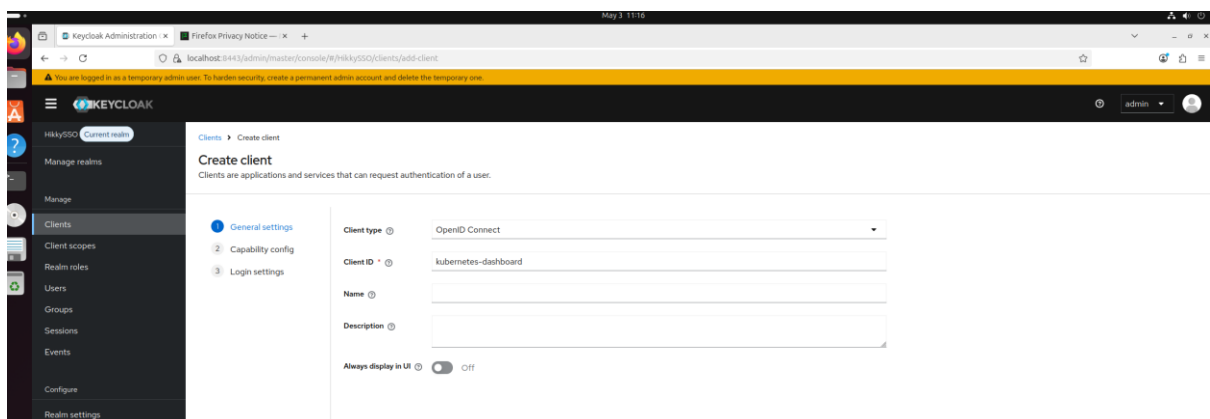
SSO

6. Creating Users

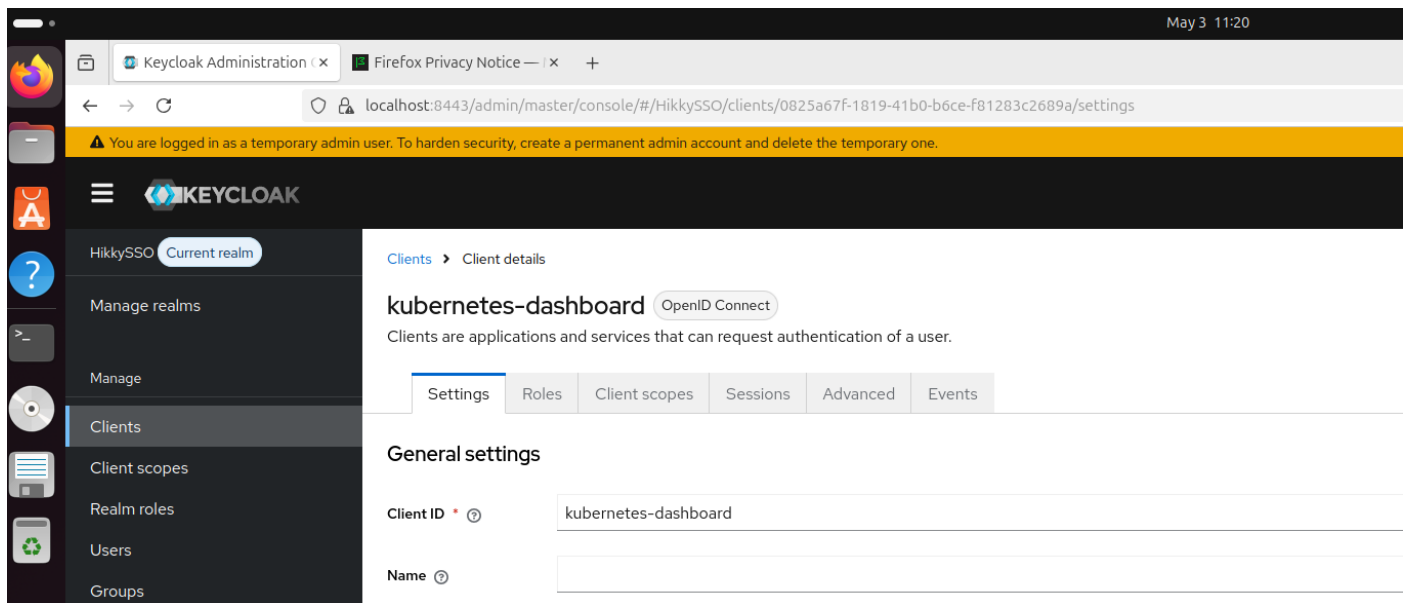
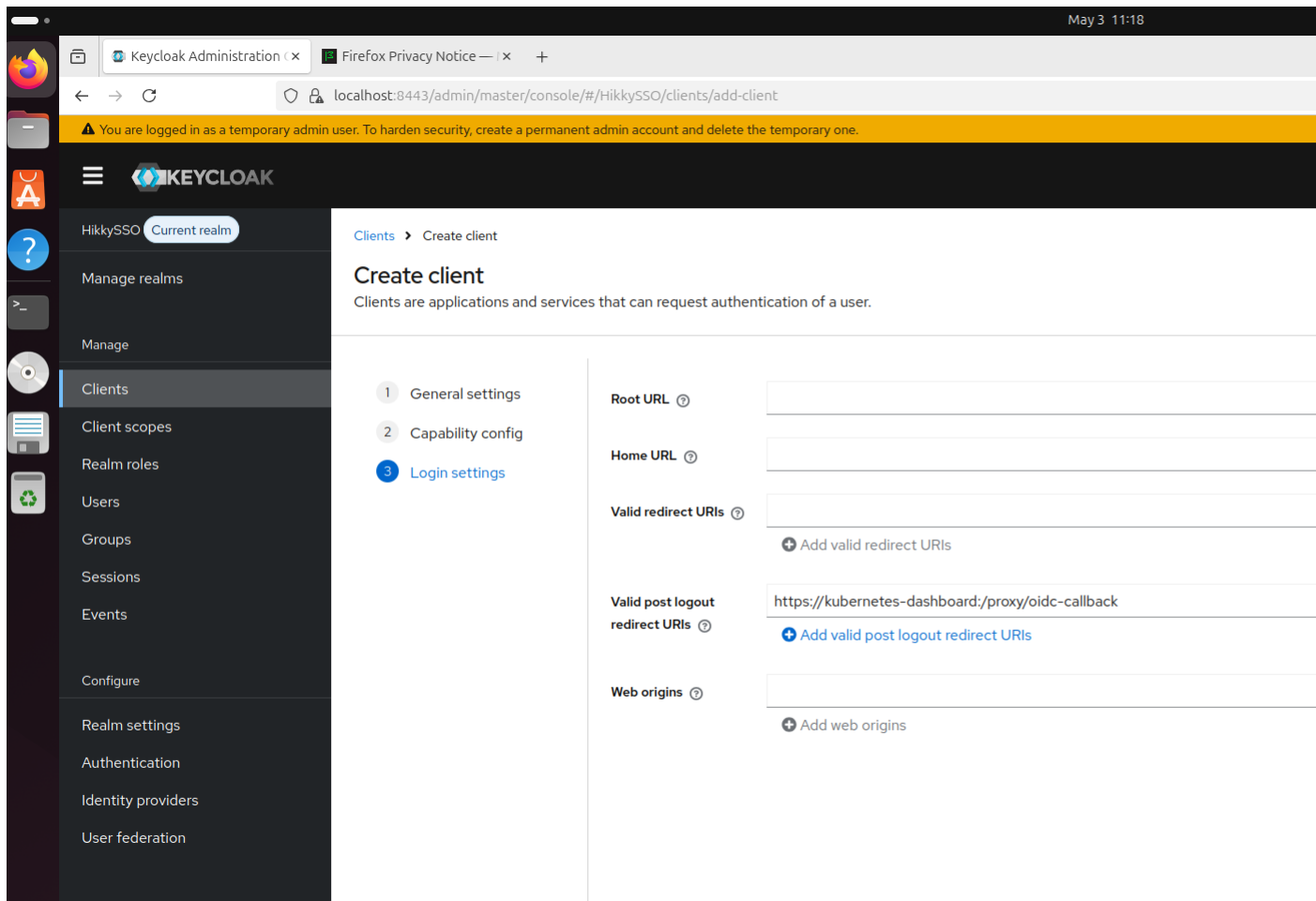


The screenshot shows the Keycloak administration interface. The left sidebar contains a menu with options like 'Manage realms', 'Clients', 'Groups', 'Sessions', 'Events', 'Configure', 'Realm settings', 'Authentication', 'Identity providers', and 'User federation'. The main content area is titled 'Users > User details' and shows the details for a user named 'hikmatabdul'. The user is currently 'Enabled'. The 'Details' tab is selected, showing fields for 'ID' (023272b2-d83e-4ef9-9f7e-7f998a9fb475), 'Created at' (5/3/2026, 11:11:27 AM), 'Required user actions' (Select action), and 'Email verified' (Off). Below this, the 'General' tab is selected, showing fields for 'Username' (hikmatabdul), 'Email', 'First name', and 'Last name'. A 'Jump to section' link is also visible.

7. Creating OIDC Client



The screenshot shows the Keycloak administration interface. The left sidebar is the same as in the previous screenshot. The main content area is titled 'Clients > Create client' and shows the 'Create client' form. The form has three tabs: 'General settings', 'Capability config', and 'Login settings'. The 'General settings' tab is selected, showing fields for 'Client type' (OpenID Connect), 'Client ID' (kubernetes-dashboard), 'Name', 'Description', and 'Always display in UI' (Off).



8. Preparing Kubernetes MikroK8s + Dashboard

```

hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ sudo snap install microk8s --classic --channel=1.30/stable
sudo usermod -aG microk8s $USER
newgrp microk8s
[sudo] password for hikmat:
Sorry, try again.
[sudo] password for hikmat:
Ensure prerequisites for "microk8s" are available
Download snap "core20" (2769) from channel "stable"
Download snap "core20" (2769) from channel "stable"
Download snap "core20" (2769) from channel "stable"
microk8s (1.30/stable) v1.30.14 from Canonical✓ installed
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$

```

```

hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ microk8s enable dashboard
microk8s kubectl get pods -A | grep dashboard
Infer repository core for addon dashboard
Enabling Kubernetes Dashboard
Infer repository core for addon metrics-server
Enabling Metrics-Server
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created
clusterrolebinding.rbac.authorization.k8s.io/microk8s-admin created
Metrics-Server is enabled
Applying manifest
serviceaccount/kubernetes-dashboard created
service/kubernetes-dashboard created
secret/kubernetes-dashboard-certs created
secret/kubernetes-dashboard-csrf created
secret/kubernetes-dashboard-key-holder created
configmap/kubernetes-dashboard-settings created
role.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrole.rbac.authorization.k8s.io/kubernetes-dashboard created
rolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
deployment.apps/kubernetes-dashboard created
service/dashboard-metrics-scraper created
deployment.apps/dashboard-metrics-scraper created
secret/microk8s-dashboard-token created

If RBAC is not enabled access the dashboard using the token retrieved with:

microk8s kubectl describe secret -n kube-system microk8s-dashboard-token

Use this token in the https login UI of the kubernetes-dashboard service.

In an RBAC enabled setup (microk8s enable RBAC) you need to create a user with restricted
permissions as shown in:
https://github.com/kubernetes/dashboard/blob/master/docs/user/access-control/creating-sample-user.md

kube-system dashboard-metrics-scraper-795895d745-mddv8 0/1 ContainerCreating 0 18s
kube-system kubernetes-dashboard-6796797fb5-x77bz 0/1 ContainerCreating 0 21s
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$

```

9. Make Kubernetes API Server Trust Keycloak (OIDC)

```

hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ sudo snap install kubectl --classic
kubectl 1.35.4 from Canonical✓ installed
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$ sudo nano /var/snap/microk8s/current/args/kube-apiserver
hikmat@hikmat-VMware-Virtual-Platform:~/keycloak-26.3.4$

```

```
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4
$ nano /var/snap/microk8s/current/args/kube-apiserver
--cert-dir=$(SNAP_DATA)/certs
--service-cluster-ip-range=10.152.183.0/24
--authorization-mode=AlwaysAllow
--service-account-key-file=$(SNAP_DATA)/certs/serviceaccount.key
--client-ca-file=$(SNAP_DATA)/certs/ca.crt
--tls-cert-file=$(SNAP_DATA)/certs/server.crt
--tls-private-key-file=$(SNAP_DATA)/certs/server.key
--tls-cipher-suites=TLS_AES_128_GCM_SHA256,TLS_AES_256_GCM_SHA384,TLS_CHACHA20_POLY1305_SHA256,TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA,TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA,TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
--kubelet-client-certificate=$(SNAP_DATA)/certs/apiserver-kubelet-client.crt
--kubelet-client-key=$(SNAP_DATA)/certs/apiserver-kubelet-client.key
--secure-port=6443
--etcd-servers="unix://$(SNAP_DATA)/var/kubernetes/backend/kine.sock:12379"
--allow-privileged=true
--service-account-issuer="https://kubernetes.default.svc"
--service-account-signing-key-file=$(SNAP_DATA)/certs/serviceaccount.key
--event-ttl=5m
--profiling=false

# Enable the aggregation layer
--requestheader-client-ca-file=$(SNAP_DATA)/certs/front-proxy-ca.crt
--requestheader-allowed-names=front-proxy-client
--requestheader-extra-headers-prefix=X-Remote-Extra-
--requestheader-group-headers=X-Remote-Group
--requestheader-username-headers=X-Remote-User
--proxy-client-cert-file=$(SNAP_DATA)/certs/front-proxy-client.crt
--proxy-client-key-file=$(SNAP_DATA)/certs/front-proxy-client.key
# Enable the admission layer
--enable-admission-plugins=EventedAdmissionLifecycleController
--admission-control-config-file=$(SNAP_DATA)/args/admission-control-config-file.yaml
--kubelet-certificate-authority=$(SNAP_DATA)/certs/ca.crt
--kubelet-preferred-address-types=InternalIP,Hostname,InternalDNS,ExternalDNS,ExternalIP
--oidc-issuer-url=https://localhost:8443/realms/Hikky550
--oidc-client-id=kubernetes-dashboard
--oidc-username-claim=email
--oidc-groups-claim=groups
--oidc-ca-file=/usr/local/share/ca-certificates/keycloak.crt
```

```
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4$ sudo nano /var/snap/microk8s/current/args/kube-apiserver
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4$ sudo snap restart microk8s
2026-05-03T11:50:21+03:00 INFO Waiting for "snap.microk8s.daemon-containerd.service" to stop.
2026-05-03T11:50:41+03:00 INFO Waiting for "snap.microk8s.daemon-k8s-dqlite.service" to stop.
Restarted.
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4$
```

10. Patch the Dashboard to Enable OIDC Authentication

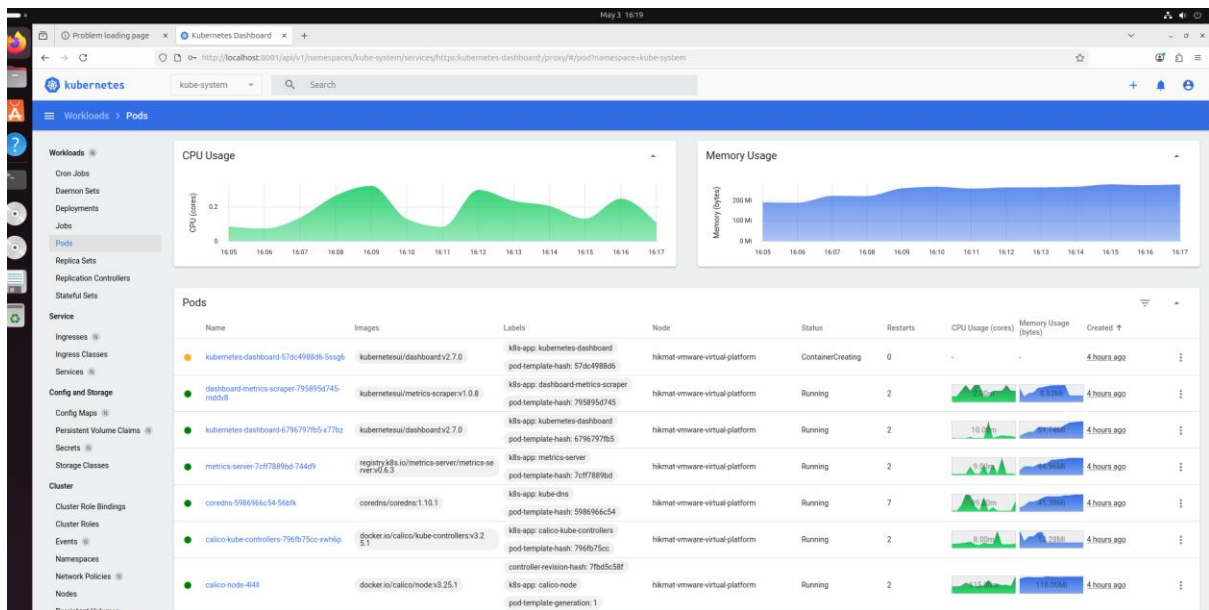
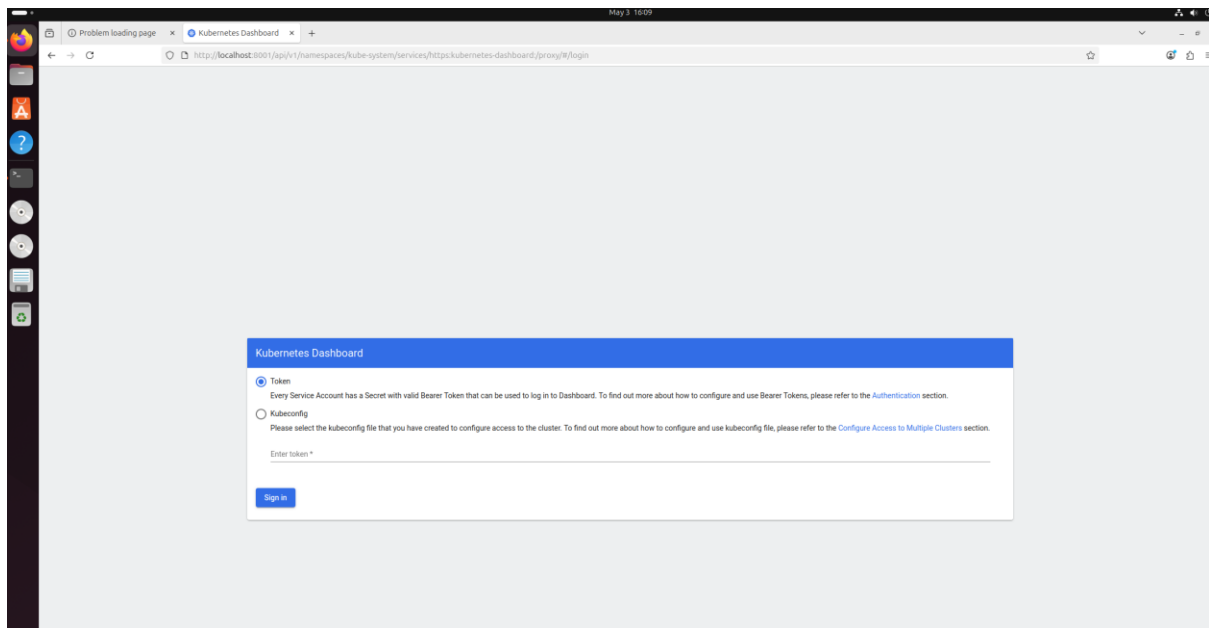
```
hikmat@hikmat-VMware-Virtual-Platform: ~/keycloak-26.3.4$ microk8s kubectl -n kube-system patch deploy kubernetes-dashboard \
--type=json' -p='[
  {
    "op": "add", "path": "/spec/template/spec/volumes/-", "value": {
      "name": "keycloak-ca",
      "configMap": {
        "name": "keycloak-ca"
      }
    }
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/volumeMounts/-", "value": {
      "name": "keycloak-ca",
      "mountPath": "/etc/keycloak",
      "readOnly": true
    }
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--authentication-mode=oidc",
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--oidc-issuer-url=https://localhost:8443/realms/SS0demo",
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--oidc-client-id=kubernetes-dashboard",
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--oidc-username-claim=email",
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--oidc-groups-claim=groups",
  },
  {
    "op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--oidc-ca-file=/etc/keycloak/ca.crt"
  }
]'
```

11. Confirming the OIDC Authentication

```
hikmat@hikmat-VMware-Virtual-Platform:~$ microk8s kubectl -n kube-system get deploy kubernetes-dashboard \
-o yaml | grep -A2 'authentication-mode'
- --authentication-mode=oidc
- --oidc-issuer-url=https://localhost:8443/realms/SS0demo
- --oidc-client-id=kubernetes-dashboard

hikmat@hikmat-VMware-Virtual-Platform:~$ microk8s kubectl get pods -n kube-system | grep dashboard
dashboard-metrics-scraper-795895d745-mddv8      1/1      Running      2 (23m ago)    4h55m
kubernetes-dashboard-57dc4988d6-5ssg6          0/1      ContainerCreating      0              4h31m
kubernetes-dashboard-6796797fb5-x77bz          1/1      Running      2 (23m ago)    4h55m
hikmat@hikmat-VMware-Virtual-Platform:~$
```

12. Testing the Kubernetes dashboard in Firefox



VAULT

1. Adding HashiCorp Repository

```

hikmat@hikmat-VMware-Virtual-Platform:~$ sudo apt update
sudo apt install -y wget gpg

wget -O- https://apt.releases.hashicorp.com/gpg | \
  sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg

echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] \
  https://apt.releases.hashicorp.com $(lsb_release -cs) main" | \
  sudo tee /etc/apt/sources.list.d/hashicorp.list
[sudo] password for hikmat:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
gpg is already the newest version (2.4.4-2ubuntu17.4).
gpg set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
--2026-05-03 16:41:12-- https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com (apt.releases.hashicorp.com)... 18.165.140.52, 18.165.140.122, 18.165.140.50, ...
Connecting to apt.releases.hashicorp.com (apt.releases.hashicorp.com)[18.165.140.52]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3980 (3.9K) [binary/octet-stream]
Saving to: 'STDOUT'

-
100%[=====]

2026-05-03 16:41:12 (693 MB/s) - written to stdout [3980/3980]

deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com noble main
hikmat@hikmat-VMware-Virtual-Platform:~$

```

2. Installing Vault


```
May 3 16:51
hikmat@hikmat-VMware-Virtual-Platform: ~
hikmat@hikmat-VMware-Virtual-Platform:~$ vault server -config/home/hikmat/vault/config/vault.hcl

==> Vault server configuration:

Administrative Namespace:
  Api Address: http://127.0.0.1:8200
  Cgo: disabled
  Cluster Address: https://127.0.0.1:8201
  Environment Variables: COLORTERM, DBUS_SESSION_BUS_ADDRESS, DEBUGINFO_URLS, DESKTOP_SESSION, DISPLAY, GDMSESSION, GNOME_DESKTOP_SESSION_ID, GNOME_KEYRING_CONTROL, GNOME_SETUP_DISPLAY, GNOME_SHELL_SESSION_MODE, GNOME_TERMINAL_SCREEN, GNOME_TERMINAL_SERVICE, GPG_AGENT_INFO, GTK_MODULES, HOME, IM_CONFIG_PHASE, LANG, LESSCLOSE, LESSOPEN, LOGNAME, LS_COLORS, MEMORY_PRESSURE_WATCH, MEMORY_PRESSURE_WRITE, PATH, PWD, QT_ACCESSIBILITY, QT_IM_MODULE, SESSION_MANAGER, SHELL, SHLVL, SSH_AUTH_SOCK, SYSTEMD_EXEC_PID, TERM, USER, USERNAME, VTE_VERSION, WAYLAND_DISPLAY, XAUTHORITY, XDG_CONFIG_DIRS, XDG_CURRENT_DESKTOP, XDG_DATA_DIRS, XDG_MENU_PREFIX, XDG_RUNTIME_DIR, XDG_SESSION_CLASS, XDG_SESSION_DESKTOP, XDG_SESSION_TYPE, XMODIFIERS, _
  Go Version: go1.26.2
  Listener 1: tcp (addr: "0.0.0.0:8200", cluster address: "0.0.0.0:8201", disable_request_limiter: "false", max_json_array_element_count: "10000", max_json_depth: "300", max_json_object_entry_count: "10000", max_json_string_value_length: "1048576", max_request_duration: "1m30s", max_request_size: "33554432", tls: "disabled")
  Log Level:
  Mock: supported: true, enabled: true
  Recovery Mode: false
  Storage: file
  Version: Vault v2.0.0, built 2026-04-13T18:49:01Z
  Version Sha: c1ceddf8a6ee32358bd5f8c37e771388cb951

==> Vault server started! Log data will stream in below:

2026-05-03T16:51:07.080+0300 [INFO] proxy environment: http_proxy="" https_proxy="" no_proxy=""
2026-05-03T16:51:07.089+0300 [INFO] incrementing seal generation: generations1
2026-05-03T16:51:09.946+0300 [INFO] core: Initializing version history cache for core
2026-05-03T16:51:09.952+0300 [INFO] events: Starting event system
```

5. Setting Vault Address

```
hikmat@hikmat-VMware-Virtual-Platform:~$ export VAULT_ADDR='http://127.0.0.1:8200'
hikmat@hikmat-VMware-Virtual-Platform:~$ vault operator init
Unseal Key 1: eMLi6xbBNHoC+eoh9CTu7ZN8JI2zMSaZeYeMECm5oiwD
Unseal Key 2: EAgW378FXhZ8Cyc7kLCxWtcVp1MKBZP+c2wCJUpLU6Am
Unseal Key 3: vrehwZpeSbn3rXB+xZxt9PCfaE8Fng+NYc+j9Qq092qS
Unseal Key 4: rOAV34hJRnXLJnD9m7IquQMlyGiAmA57nWazz/Fv/50a
Unseal Key 5: xLaNgAHF130GGs8qUT2ht0v8JVMuZowAxURPMwFh3hzO

Initial Root Token: hvs.03SR1D66nzYECUDazRwu10xa

Vault initialized with 5 key shares and a key threshold of 3. Please securely
distribute the key shares printed above. When the Vault is re-sealed,
restarted, or stopped, you must supply at least 3 of these keys to unseal it
before it can start servicing requests.

Vault does not store the generated root key. Without at least 3 keys to
reconstruct the root key, Vault will remain permanently sealed!

It is possible to generate new unseal keys, provided you have a quorum of
existing unseal keys shares. See "vault operator rekey" for more information.
hikmat@hikmat-VMware-Virtual-Platform:~$
```

6. I Unsealed the Vault

```

hikmat@hikmat-VMware-Virtual-Platform:~$ vault operator unseal
# paste Unseal Key 1 when prompted

vault operator unseal
# paste Unseal Key 2 when prompted

vault operator unseal
# paste Unseal Key 3 when prompted
Unseal Key (will be hidden):
Key          Value
---          -
Seal Type    shamir
Initialized  true
Sealed       true
Total Shares 5
Threshold    3
Unseal Progress 1/3
Unseal Nonce  af9592d6-bff0-f8a0-5aa0-e7e43ffd8ddf
Version      2.0.0
Build Date   2026-04-13T18:49:01Z
Storage Type  file
HA Enabled    false
Unseal Key (will be hidden):
Key          Value
---          -
Seal Type    shamir
Initialized  true
Sealed       true
Total Shares 5
Threshold    3
Unseal Progress 2/3
Unseal Nonce  af9592d6-bff0-f8a0-5aa0-e7e43ffd8ddf
Version      2.0.0
Build Date   2026-04-13T18:49:01Z
Storage Type  file
HA Enabled    false
Unseal Key (will be hidden):
Key          Value
---          -
Seal Type    shamir
Initialized  true
Sealed       false
Total Shares 5
Threshold    3
Version      2.0.0
Build Date   2026-04-13T18:49:01Z
Storage Type  file
Cluster Name  vault-cluster-014f5020
Cluster ID    26b6d826-ec9e-a98a-6162-af257947a968
HA Enabled    false

```

7. Log into Vault

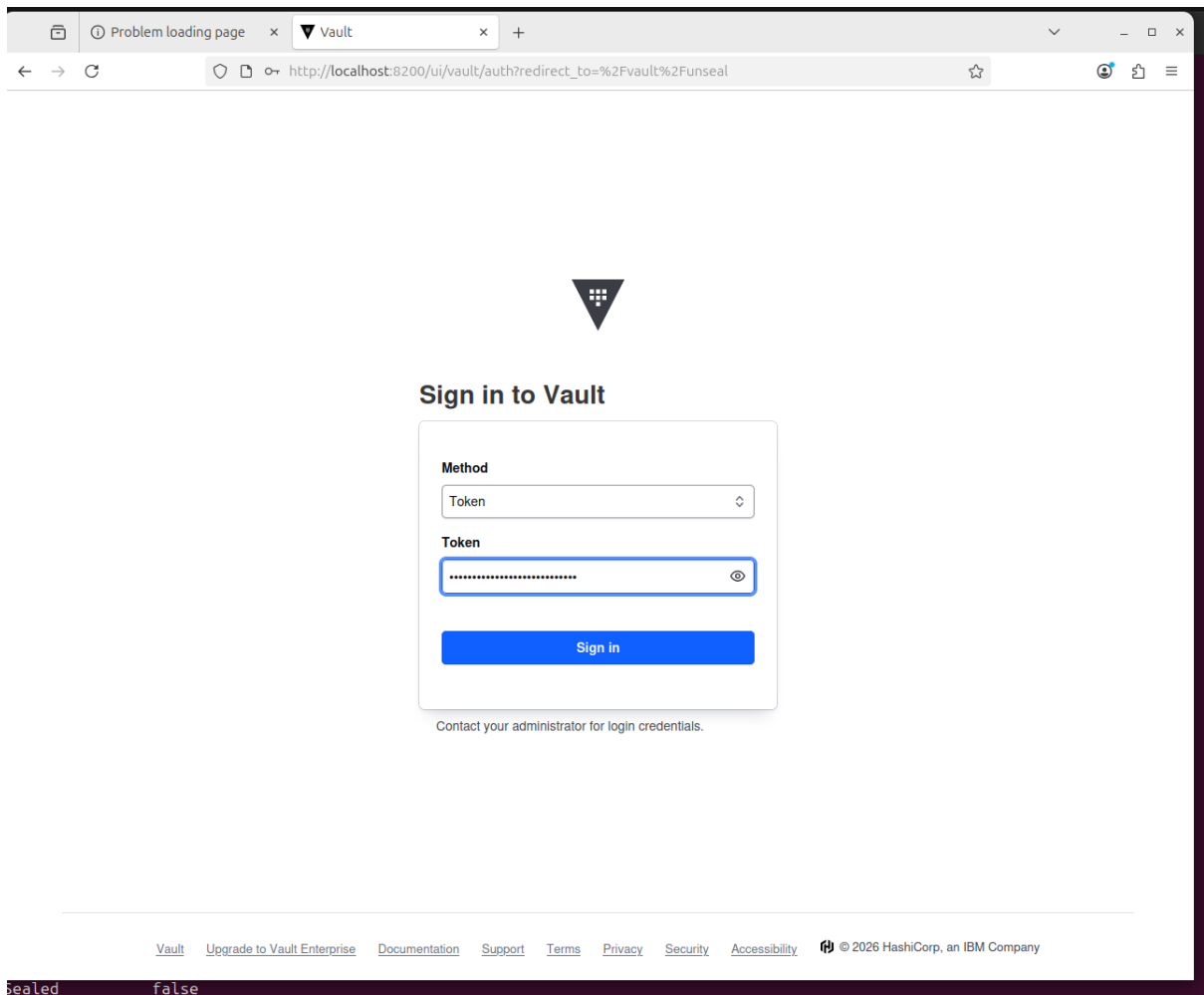
```

hikmat@hikmat-VMware-Virtual-Platform:~$ vault login
# paste your Initial Root Token when prompted
Token (will be hidden):
Success! You are now authenticated. The token information displayed below
is already stored in the token helper. You do NOT need to run "vault login"
again. Future Vault requests will automatically use this token.

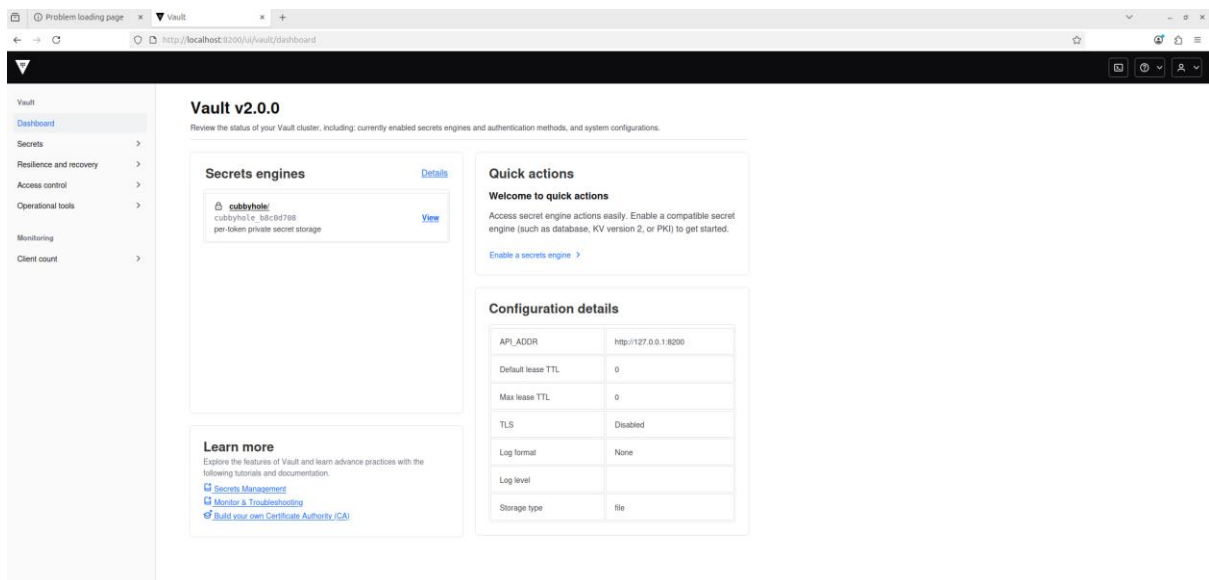
Key          Value
---          -
token        hvs.03SR1D66nzYECUDazRwu10xa
token_accessor bEsDyiIg02i6rX4s49F9XHHi
token_duration ∞
token_renewable false
token_policies ["root"]
identity_policies []
policies       ["root"]
hikmat@hikmat-VMware-Virtual-Platform:~$ █

```

8. Opened Vault UI



sealed false

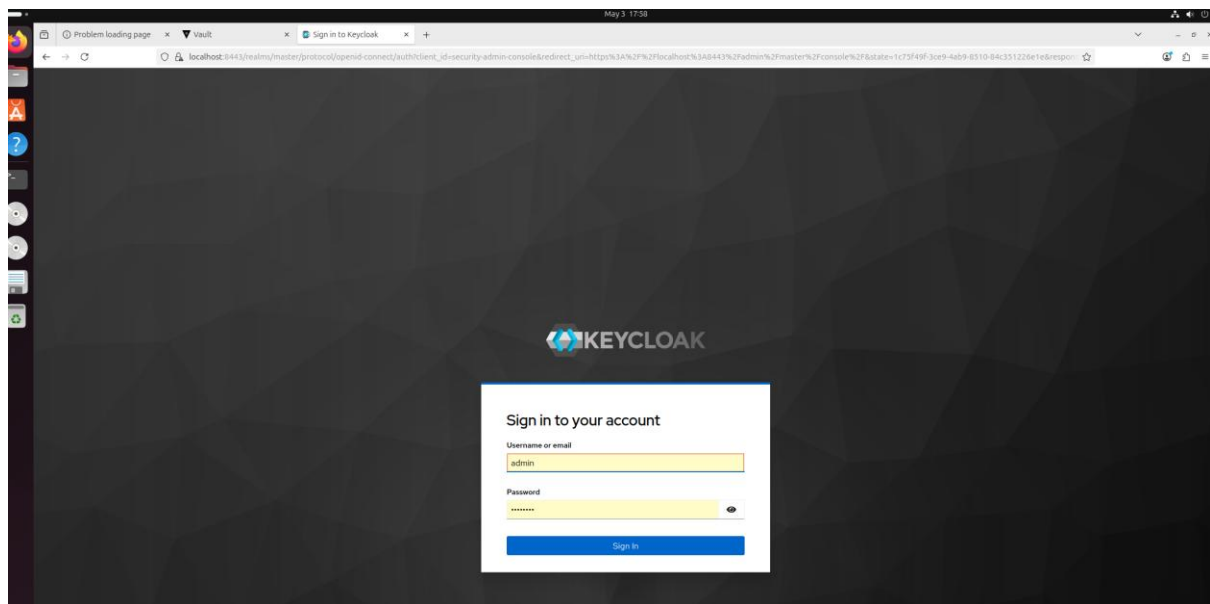


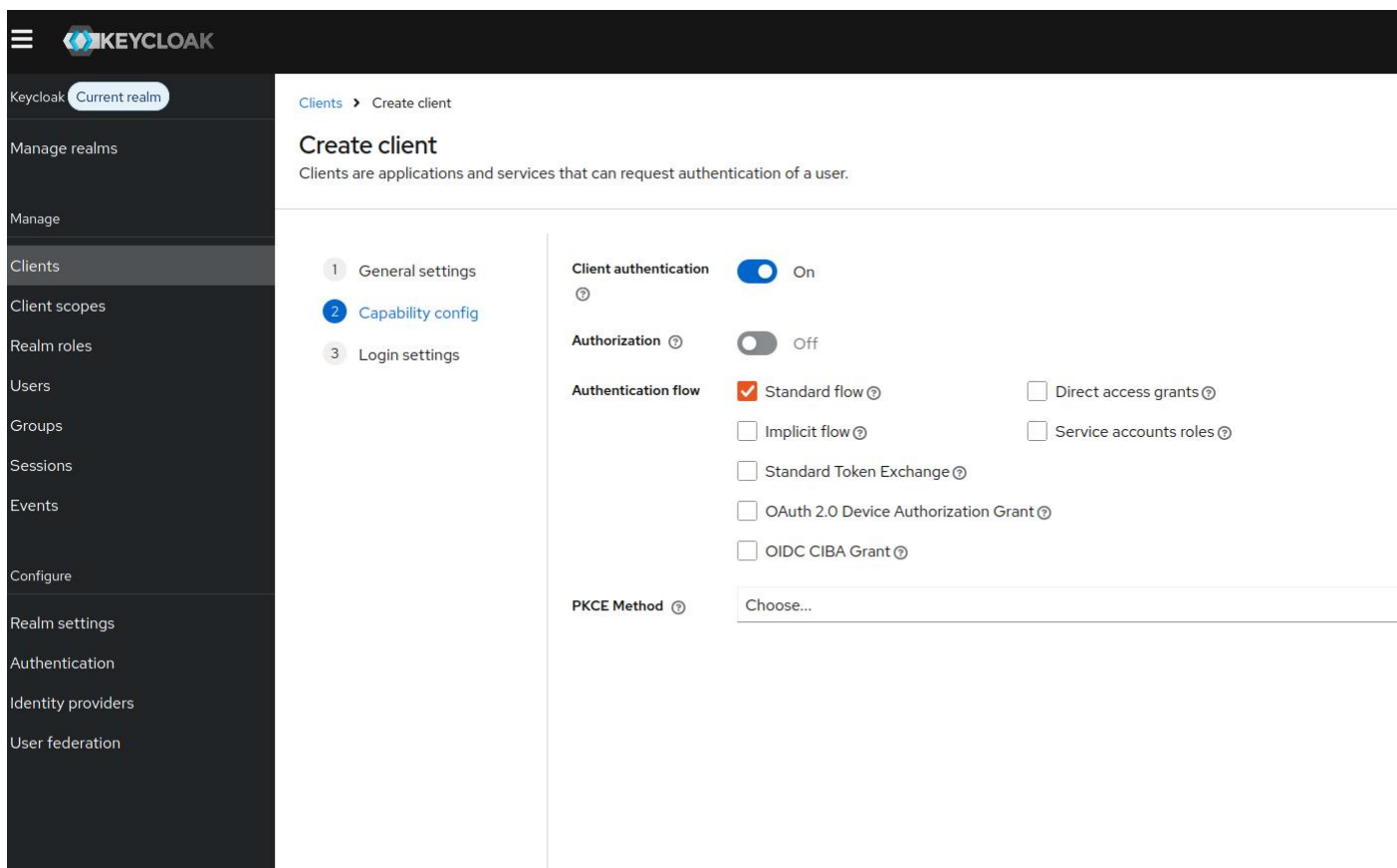
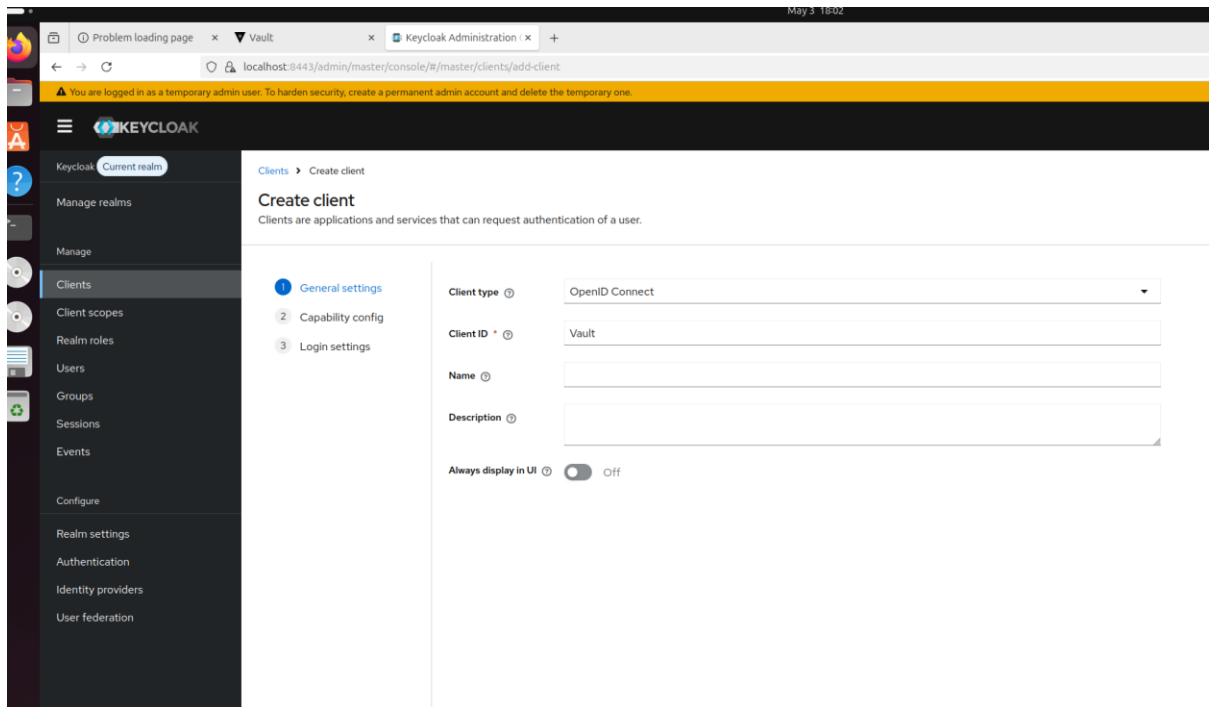
9. Logged In via Terminal

```
hikmat@hikmat-VMware-Virtual-Platform:~$ export VAULT_ADDR='http://127.0.0.1:8200'
vault login
# paste your Root Token when prompted
Token (will be hidden):
Success! You are now authenticated. The token information displayed below
is already stored in the token helper. You do NOT need to run "vault login"
again. Future Vault requests will automatically use this token.

Key          Value
---          -
token        hvs.03SR1D66nzYECUDazRwu10xa
token_accessor bEsDyiIg02i6rX4s49F9XHHi
token_duration ∞
token_renewable false
token_policies ["root"]
identity_policies []
policies      ["root"]
hikmat@hikmat-VMware-Virtual-Platform:~$
```

10. Created Vault Client in Keycloak





[Clients](#) > Create client

Create client

Clients are applications and services that can request authentication of a user.

1 General settings

2 Capability config

3 Login settings

Root URL ⓘ

Home URL ⓘ

Valid redirect URIs ⓘ

http://localhost:8200/ui/vault/auth/oidc/oidc/callback

http://localhost:8250/oidc/callback

+ Add valid redirect URIs

Valid post logout redirect URIs ⓘ

+ Add valid post logout redirect URIs

Web origins ⓘ

+ Add web origins

May 3 18:08

Problem loading page x Vault x Keycloak Administration x +

localhost:8443/admin/master/console/#/master/clients/35cd66e6-9a4e-453a-a889-6261ec5a18e7/credentials

You are logged in as a temporary admin user. To harden security, create a permanent admin account and delete the temporary one.

KEYCLOAK

Keycloak Current realm

Manage realms

Manage

Clients

Client scopes

Realm roles

Users

Groups

Sessions

Events

Configure

Realm settings

Authentication

Identity providers

User federation

Clients > Client details

Vault ⓘ OpenID Connect ⓘ

Clients are applications and services that can request authentication of a user.

Settings Keys Credentials Roles Client scopes Sessions Advanced Events

Client Authenticator ⓘ Client Id and Secret ⓘ

Save

Client Secret ⓘ ⓘ ⓘ Regenerate

Registration access token ⓘ ⓘ Regenerate

11. Configured and created the OIDC Role

```

hikmat@hikmat-VMware-Virtual-Platform:~$ export VAULT_ADDR='http://127.0.0.1:8200'

# Login to Vault first
vault login
# paste your Root Token
Token (will be hidden):
Success! You are now authenticated. The token information displayed below
is already stored in the token helper. You do NOT need to run "vault login"
again. Future Vault requests will automatically use this token.

Key                Value
---                -
token              hvs.03SR1D66nzYECUDazRwu10xa
token_accessor     bEsDyiIg02i6rX4s49F9XHHI
token_duration     ∞
token_renewable    false
token_policies     ["root"]
identity_policies  []
policies           ["root"]
hikmat@hikmat-VMware-Virtual-Platform:~$ vault auth enable oidc
Success! Enabled oidc auth method at: oidc/
hikmat@hikmat-VMware-Virtual-Platform:~$ vault write auth/oidc/config \
  oidc_discovery_url="https://localhost:8443/realms/master" \
  oidc_client_id="vault" \
  oidc_client_secret="zSCDESmZNCDym2pbWLFd8WndC23u0GNn" \
  default_role="default" \
  oidc_discovery_ca_pem=@/usr/local/share/ca-certificates/keycloak.com
Failed to parse K=V data: invalid key/value pair "oidc_discovery_ca_pem=@/usr/local/share/ca-certif
hikmat@hikmat-VMware-Virtual-Platform:~$ vault write auth/oidc/config \
  oidc_discovery_url="https://localhost:8443/realms/master" \
  oidc_client_id="vault" \
  oidc_client_secret="zSCDESmZNCDym2pbWLFd8WndC23u0GNn" \
  default_role="default" \
  oidc_discovery_ca_pem=@/usr/local/share/ca-certificates/keycloak.crt

Success! Data written to: auth/oidc/config
hikmat@hikmat-VMware-Virtual-Platform:~$ vault write auth/oidc/role/default \
  bound_audiences="vault" \
  allowed_redirect_uris="http://localhost:8200/ui/vault/auth/oidc/oidc/callback" \
  allowed_redirect_uris="http://localhost:8250/oidc/callback" \
  user_claim="sub" \
  policies="default"
Success! Data written to: auth/oidc/role/default
hikmat@hikmat-VMware-Virtual-Platform:~$ █

```

12. Test Login via Vault UI



Sign in to Vault

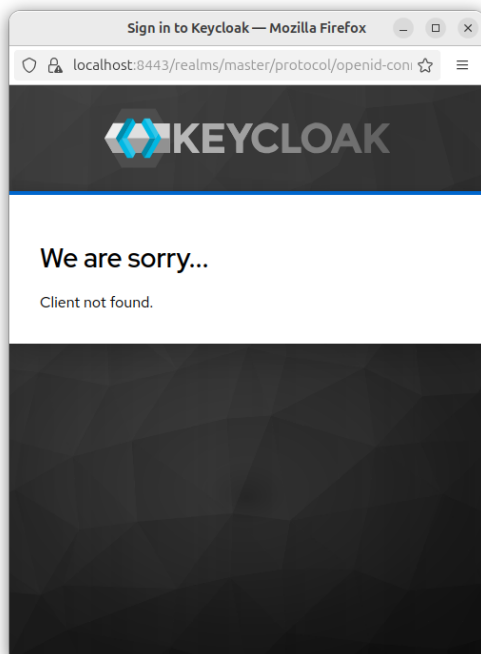
Method

Role
Vault will use the default role to sign in if this field is left blank.

[Advanced settings](#)

[Sign in with OIDC Provider](#)

Contact your administrator for login credentials.



Sign in to Vault

Method

Role
Vault will use the default role to sign in if this field is left blank.

[Advanced settings](#)

[Sign in with OIDC Provider](#)

Contact your administrator for login credentials.